3 Proof of the main results

3.1 Annealed free energy and coefficient asymptotics

To identify the deterministic coefficient $A_N(\beta)$ in (**??**), we use the annealed free-energy limit from [**?**], which holds at every fixed inverse temperature. Convexity then gives locally uniform convergence of the first derivatives.

The following is a standard convergence theorem for convex functions; see [**?**, Theorems 10.8 and 25.7].

Lemma 3.1. *Let $I \subset \mathbb{R}$ be open. Suppose that $h_N : I \to \mathbb{R}$ are finite, differentiable, convex functions and that $h_N \to h$ pointwise, where $h \in C^1(I)$. Then locally uniformly on I,*

$$h_N \to h \qquad \text{and} \quad h_N' \to h'.$$

Proof. Fix $[a, b] \subset I$. For $x \in I$ and sufficiently small $s > 0$, convexity gives

$$\frac{h_N(x) - h_N(x - s)}{s} \le h_N'(x) \le \frac{h_N(x + s) - h_N(x)}{s}.$$

Taking lim inf and lim sup as $N \to \infty$, using pointwise convergence, and then letting $s \downarrow 0$, gives

$$h'(x) \le \liminf_N h_N'(x) \le \limsup_N h_N'(x) \le h'(x).$$

Thus $h_N'(x) \to h'(x)$ pointwise.

Each h_N' is nondecreasing. Given $\varepsilon > 0$, by uniform continuity of h' on $[a, b]$, choose a partition $a = x_0 < x_1 < \cdots < x_m = b$ such that, for every $i = 0, \dots, m - 1$,

$$\sup_{x,y \in [x_i, x_{i+1}]} |h'(x) - h'(y)| < \varepsilon.$$

Pointwise convergence at the finitely many x_i gives, for all sufficiently large N and every $i = 0, \dots, m$,

$$|h_N'(x_i) - h'(x_i)| < \varepsilon.$$

For $x \in [x_i, x_{i+1}]$, monotonicity of h_N' gives

$$h_N'(x_i) \le h_N'(x) \le h_N'(x_{i+1}),$$

and hence

$$|h_N'(x) - h'(x)| < 2\varepsilon.$$

Therefore, $h_N' \to h'$ uniformly on $[a, b]$.

By the mean value theorem, we also have that $h_N \to h$ uniformly on $[a, b]$. □

Proposition 3.2. *There exists an even convex function $\ell_q \in C^1(\mathbb{R})$ such that, locally uniformly for $\beta \in \mathbb{R}$,*

$$\frac{1}{N} \log \mathbb{E} Z_N(\beta) \longrightarrow \ell_q(\beta), \tag{1}$$

$$\frac{1}{N} \partial_\beta \log \mathbb{E} Z_N(\beta) \longrightarrow \ell_q'(\beta). \tag{2}$$

Moreover, for every $\beta > 0$,

$$0 < \ell_q'(\beta) \le \frac{\beta}{q}, \qquad 0 \le \frac{\beta}{q} - \ell_q'(\beta) \le \frac{\beta^3}{3}, \tag{3}$$

and, for every $\beta \in \mathbb{R}$,

$$0 \le \frac{\beta^2}{2q} - \ell_q(\beta) \le \frac{\beta^4}{12}. \tag{4}$$

In particular, as $\beta \to 0$,

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(|\beta|^3). \tag{5}$$

Furthermore, $\beta \ell_q'(\beta) > 0$ *for every* $\beta \neq 0$.

Proof. We first record the normalization used in [**?**]. Set

$$H_N^{\mathrm{He}} := 2^{-q/2} H_N, \qquad p_{N,q}^{\mathrm{He}}(s) := \frac{1}{N} \log \mathbb{E} \operatorname{tr} e^{s H_N^{\mathrm{He}}}.$$

Note that the paper also uses the normalized trace. Since G and $-G$ have the same law and $H_N(-G) = -H_N(G)$, for every $\beta \in \mathbb{R}$,

$$\frac{1}{N} \log \mathbb{E} Z_N(\beta) = \frac{1}{N} \log \mathbb{E} \operatorname{tr} e^{|\beta| H_N} = p_{N,q}^{\mathrm{He}}(2^{q/2}|\beta|). \tag{6}$$

Writing $p_q := p_{\mathrm{SD},q}$ for the Schwinger–Dyson pressure in [**?**], [**?**, Theorem 1.1 and Proposition 3.3] give

$$p_{N,q}^{\mathrm{He}}(s) \longrightarrow p_q(s) \quad (s > 0), \qquad p_q \in C^1((0,\infty)), \qquad \lim_{s \downarrow 0} p_q(s) = 0.$$

Define

$$\ell_q(\beta) := \begin{cases} p_q(2^{q/2}|\beta|), & \beta \neq 0, \\ 0, & \beta = 0. \end{cases}$$

Then ℓ_q is even and continuous on $\mathbb{R}$, and (6), together with $Z_N(0) = 1$, proves pointwise convergence in (1).

We next use the derivative formula in [**?**, Proposition 3.3]: for any $s > 0$,

$$p_q'(s) = \frac{s}{q} \int_0^1 G_{s,q}(u)^q \, \mathrm{d}u. \tag{7}$$

By the positive Lehmann representation and [**?**, Theorem 2.5], there is a probability measure ν_s on $[0,\infty)$ such that

$$G_{s,q}(u) = \int_0^\infty \frac{\cosh(x(1/2-u))}{2\cosh(x/2)} \, \nu_s(\mathrm{d}x), \qquad \int_0^\infty x^2 \, \nu_s(\mathrm{d}x) = 2^{2-q} s^2.$$

For $\beta > 0$, put

$$K_\beta(u) := 2 G_{2^{q/2}\beta,q}(u).$$

The chain rule and (7) yield

$$\ell_q'(\beta) = 2^{q/2} p_q'(2^{q/2}\beta) = \frac{2^q \beta}{q} \int_0^1 G_{2^{q/2}\beta,q}(u)^q \, \mathrm{d}u = \frac{\beta}{q} \int_0^1 K_\beta(u)^q \, \mathrm{d}u. \tag{8}$$

Moreover,

$$K_\beta(u) = \int_0^\infty \frac{\cosh(x(1/2-u))}{\cosh(x/2)} \, \nu_{2^{q/2}\beta}(\mathrm{d}x), \qquad \int_0^\infty x^2 \, \nu_{2^{q/2}\beta}(\mathrm{d}x) = 4\beta^2. \tag{9}$$

For $x \ge 0$ and $u \in [0,1]$, the inequality $\tanh v \le v$ for $v \ge 0$ gives

$$\begin{aligned} 0 &\le \log\cosh(x/2) - \log\cosh(x(1/2-u)) \\ &= \int_{x|1/2-u|}^{x/2} \tanh v \, \mathrm{d}v \\ &\le \frac{1}{2}\left(\frac{x^2}{4} - x^2(1/2-u)^2 \right) = \frac{x^2}{2} u(1-u). \end{aligned}$$

Using $1 - e^{-a} \le a$ for $a \ge 0$, we obtain

$$0 \le 1 - \frac{\cosh(x(1/2-u))}{\cosh(x/2)} \le 1 - e^{-x^2 u(1-u)/2} \le \frac{x^2}{2} u(1-u).$$

Integrating with respect to $\nu_{2^{q/2}\beta}$ in (9) therefore gives

$$0 < K_\beta(u) \le 1, \qquad 0 \le 1 - K_\beta(u) \le 2\beta^2 u(1-u). \tag{10}$$

The first inequality and (8) imply $0 < \ell_q'(\beta) \le \beta/q$. For the remaining bound, use

$$1 - a^q = (1-a) \sum_{j=0}^{q-1} a^j \le q(1-a), \qquad 0 \le a \le 1,$$

to deduce that

$$\begin{aligned} 0 &\le \frac{\beta}{q} - \ell_q'(\beta) \\ &= \frac{\beta}{q} \int_0^1 \bigl(1 - K_\beta(u)^q\bigr)\, \mathrm{d}u \\ &\le \beta \int_0^1 \bigl(1 - K_\beta(u)\bigr)\, \mathrm{d}u \\ &\le 2\beta^3 \int_0^1 u(1-u)\, \mathrm{d}u = \frac{\beta^3}{3}. \end{aligned}$$

This proves (3). Integrating from 0 to β with $\ell_q(0) = 0$, gives

$$0 \le \frac{\beta^2}{2q} - \ell_q(\beta) = \int_0^\beta \left(\frac{s}{q} - \ell_q'(s)\right) \mathrm{d}s \le \int_0^\beta \frac{s^3}{3}\, \mathrm{d}s = \frac{\beta^4}{12}.$$

This proves (4) for $\beta > 0$. Since ℓ_q is even as we proved, we have the claim for all $\beta \in \mathbb{R}$.

The bound (4) implies

$$\lim_{\beta \to 0} \frac{\ell_q(\beta) - \ell_q(0)}{\beta} = 0.$$

Hence $\ell_q'(0) = 0$. On $\mathbb{R} \setminus \{0\}$, the derivative ℓ_q' is continuous by $p_q \in C^1((0,\infty))$ and the chain rule, and is odd by evenness of ℓ_q. By (3),

$$|\ell_q'(\beta)| \le \frac{|\beta|}{q},$$

so it is also continuous at zero. Thus $\ell_q \in C^1(\mathbb{R})$. Equations (3) and (4) give (5), and oddness of ℓ_q' gives $\beta \ell_q'(\beta) > 0$ for $\beta \ne 0$.

It remains to prove convexity and the two local-uniform convergences. Set

$$F_N(\beta) := \mathbb{E} Z_N(\beta), \qquad h_N(\beta) := \frac{1}{N} \log F_N(\beta), \qquad \overline{\nu}_N(B) := \mathbb{E}\, \mathrm{tr}\, \mathbf{1}_B(H_N).$$

Then $\overline{\nu}_N$ is a probability measure and

$$F_N(\beta) = \int_{\mathbb{R}} e^{-\beta x}\, \overline{\nu}_N(\mathrm{d}x).$$

In particular, h_N is finite and twice continuously differentiable. With the probability measure

$$\overline{\nu}_{N,\beta}(\mathrm{d}x) := \frac{e^{-\beta x}}{F_N(\beta)}\, \overline{\nu}_N(\mathrm{d}x),$$

we have

$$h_N''(\beta) = \frac{1}{N}\left(\int_{\mathbb{R}} x^2\,\overline{\nu}_{N,\beta}(\mathrm{d}x) - \left(\int_{\mathbb{R}} x\,\overline{\nu}_{N,\beta}(\mathrm{d}x)\right)^2\right) \geq 0.$$

Thus each h_N is convex, and so is its pointwise limit ℓ_q. Applying Lemma 3.1 with $I = \mathbb{R}$ proves both (1) and (2) locally uniformly on $\mathbb{R}$. □

By (**??**) and (2), locally uniformly for $\beta \in \mathbb{R}$,

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2}\ell_q'(\beta).$$

Proof of Theorem **??**. Let

$$S_N := \frac{1}{\sqrt{2M_N}} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1).$$

The classical i.i.d. central limit theorem gives $S_N \Rightarrow \xi \sim \mathcal{N}(0,1)$. By Lemma **??**,

$$Y_N(\beta) - 1 = c_N(\beta) S_N + \mathcal{R}_N(\beta), \qquad c_N(\beta) := \frac{\beta\,\partial_\beta \log \mathbb{E}Z_N(\beta)}{\sqrt{2M_N}}, \tag{11}$$

where $\mathcal{R}_N := \sum_{r \geq 3} P_r Y_N$ is orthogonal to S_N. Since $M_N \sim N^q/q!$, (2) gives

$$N^{q/2-1} c_N(\beta) \longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta).$$

Proposition **??** gives $N^{q/2-1}\mathcal{R}_N \to 0$ in L^2. This proves (**??**). Orthogonality in (11) also gives

$$\mathrm{Var}(Y_N) = c_N(\beta)^2 + \|\mathcal{R}_N(\beta)\|_2^2\,.$$

Finally, for $\beta \neq 0$, the identity $\mathrm{Var}(Y_N) = \mathrm{Var}(Z_N)/(\mathbb{E}Z_N)^2$ and the strict positivity $\beta\ell_q'(\beta) > 0$ give (**??**). □

[1]

Proof of Corollary **??**. The case $\beta = 0$ is trivial, so assume $\beta \neq 0$. Set

$$a := \frac{q}{2} - 1, \qquad W_N := \log Y_N(\beta).$$

Then

$$N^a\big(\log Z_N - \mathbb{E}\log Z_N\big) = N^a\big(W_N - (Y_N - 1)\big) + N^a(Y_N - 1) - N^a\mathbb{E}W_N.$$

By Theorem **??**,

$$N^a(Y_N - 1) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \xi \sim \mathcal{N}(0,1).$$

Moreover, $Y_N - 1 \to 0$ in probability and $N^a(Y_N - 1)$ is tight. Since whenever $|x| \leq 1/2$,

$$|\log(1+x) - x| \leq x^2,$$

it follows that

$$N^a\big(W_N - (Y_N - 1)\big) \xrightarrow{\mathrm{p}} 0.$$

Thus it remains to prove

$$\mathbb{E}W_N = o(N^{-a}).$$

[1] SN: The following is a revised proof

Write, as in (11),

$$Y_N - 1 = c_N S_N + \mathcal{R}_N,$$

where

$$c_N = \frac{\beta\,\partial_\beta \log \mathbb{E} Z_N(\beta)}{\sqrt{2M_N}} = O(N^{-a}), \quad S_N = \frac{1}{\sqrt{2M_N}} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1), \quad \|\mathcal{R}_N\|_2^2 = O(N^{-3a})$$

by Proposition **??**. We note that

$$\sup_N \mathbb{E} S_N^4 < \infty.$$

Orthogonality therefore gives

$$\mathbb{E}(Y_N - 1)^2 = O(N^{-2a}).$$

On the other hand, we estimate

$$\begin{aligned}
\mathbb{P}(Y_N < 1/2) &\le \mathbb{P}(|c_N S_N| > 1/4) + \mathbb{P}(|\mathcal{R}_N| > 1/4) \\
&\le C c_N^4 \mathbb{E} S_N^4 + C \|\mathcal{R}_N\|_2^2 \\
&= O(N^{-3a}).
\end{aligned}$$

Next we will prove that

$$Z_N(\beta; G) \ge 1.$$

Choose $x \in I$. The unitary U_x from Lemma **??** satisfies

$$U_x \psi_I U_x^* = -\psi_I,$$

and hence $\operatorname{tr} \psi_I = 0$. Therefore $\operatorname{tr} H_N(G) = 0$. Jensen's inequality applied to the eigenvalues of $H_N(G)$ now gives

$$Z_N(\beta; G) = \operatorname{tr} e^{-\beta H_N(G)} \ge e^{-\beta \operatorname{tr} H_N(G)} = 1.$$

Define

$$h(y) := y - 1 - \log y, \qquad y > 0.$$

Since $\mathbb{E} Y_N = 1$,

$$-\mathbb{E} W_N = \mathbb{E} h(Y_N).$$

On $\{Y_N \ge 1/2\}$,

$$h(Y_N) \le (Y_N - 1)^2,$$

and therefore

$$\mathbb{E}\big[h(Y_N) \mathbf{1}_{\{Y_N \ge 1/2\}}\big] = O(N^{-2a}).$$

On the other hand, if $Y_N < 1/2$, then

$$h(Y_N) \le -\log Y_N.$$

Since $Z_N \ge 1$,

$$-\log Y_N = \log \mathbb{E} Z_N - \log Z_N \le \log \mathbb{E} Z_N.$$

By Proposition 3.2, $\log \mathbb{E} Z_N = O(N)$, we have

$$\mathbb{E}\big[h(Y_N) \mathbf{1}_{\{Y_N < 1/2\}}\big] \le O(N)\, \mathbb{P}(Y_N < 1/2) = O(N^{1-3a}).$$

Since $q \ge 4$, we have $a \ge 1$, and hence $1 - 3a \le -2a$. Consequently,

$$0 \le -\mathbb{E} W_N = O(N^{-2a}) = O(N^{2-q}) = o(N^{-a}).$$

Combining the three terms above and applying Slutsky's theorem yields

$$N^{q/2-1}\big(\log Z_N(\beta) - \mathbb{E} \log Z_N(\beta)\big) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2} \beta^2 (\ell_q'(\beta))^2\right).$$

□

Lemma 3.3. [2] *Write*

$$W_N \coloneqq \log Y_N(\beta).$$

Then, for $\beta \neq 0$ and any $\lambda, t > 0$,

$$\mathbb{P}(W_N - \mathbb{E}W_N < -t) \leq \exp\left(-\frac{qt^2}{2\beta^2}\right), \tag{12}$$

$$\mathbb{E}\exp(\lambda(W_N - \mathbb{E}W_N)) \leq \exp\left(\frac{\lambda^2\beta^2}{2q}\right). \tag{13}$$

Proof. To apply the Gaussian concentration inequality, we view $Z_N(\beta; G)$ as a function of $G \coloneqq (g_I)_{I\in\mathcal{I}_N}$ and establish the Lipschitz bound for W_N. Fix $I \in \mathcal{I}_N$. By the one-variable Duhamel formula (**??**), we deduce that

$$\begin{aligned}
\partial_{g_I} Z_N &= \operatorname{tr} \partial_{g_I} e^{-\beta H_N(G)} \\
&= -\beta \operatorname{tr}\Big(\partial_{g_I} H_N(G) \cdot e^{-\beta H_N(G)}\Big) \\
&= -\beta\sigma_N \operatorname{tr}\big(e^{-\beta H_N}\psi_I\big).
\end{aligned}$$

Write $\langle\cdot\rangle_G$ for the expectation with respect to the normalized Gibbs state, i.e.,

$$\langle A\rangle_G \coloneqq \frac{\operatorname{tr}(Ae^{-\beta H_N(G)})}{Z_N(\beta; G)}.$$

Then we obtain

$$\partial_{g_I} W_N = -\beta\sigma_N\langle\psi_I\rangle_G. \tag{14}$$

Fix an arbitrary subset $S \subseteq [N]$ with $|S| = q - 1$. Set $A_i \coloneqq \psi_{S\cup\{i\}}$ for each $i \notin S$. Note that, by the relation $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$, it holds that

$$A_i^2 = \mathbb{I}, \qquad A_iA_j + A_jA_i = 0 \quad \text{if} \quad i \neq j. \tag{15}$$

This yields

$$\left(\sum_{i\notin S}\langle A_i\rangle_G A_i\right)^2 = \sum_{i\notin S}\langle A_i\rangle_G^2 A_i^2 + \sum_{\substack{i,j\notin S\\ i<j}}\langle A_i\rangle_G\langle A_j\rangle_G(A_iA_j + A_jA_i) = \sum_{i\notin S}\langle A_i\rangle_G^2\mathbb{I}.$$

If we write $\|\cdot\|$ for the operator norm, then, by (15), we obtain

$$\left\|\left(\sum_{i\notin S}\langle A_i\rangle_G A_i\right)^2\right\| = \sum_{i\notin S}\langle A_i\rangle_G^2.$$

Since A_i is self-adjoint, it follows that

$$\left\|\sum_{i\notin S}\langle A_i\rangle_G A_i\right\| = \sqrt{\sum_{i\notin S}\langle A_i\rangle_G^2}. \tag{16}$$

On the other hand, Hölder's inequality for Schatten norms yields

$$\left\langle\sum_{i\notin S}\langle A_i\rangle_G A_i\right\rangle_G \leq \left\|\sum_{i\notin S}\langle A_i\rangle_G A_i\right\|. \tag{17}$$

[2] SN: I think we can remove the blue.

Since the left-hand side equals $\sum_{i\notin S}\langle A_i\rangle_G^2$, we deduce from (16) and (17) that

$$\sum_{i\notin S}\langle A_i\rangle^2 \le 1. \tag{18}$$

For each $I \in \mathcal{I}_N$, there exist exactly q subsets $S \subseteq [N]$ satisfying $I = S \cup \{i\}$. Hence,

$$\sum_{\substack{S\subseteq[N]\\|S|=q-1}}\sum_{i\notin S}\langle\psi_{S\cup\{i\}}\rangle_G^2 = q\sum_{I\in\mathcal{I}_N}\langle\psi_I\rangle_G^2.$$

This, combined with (18), yields

$$\sum_{I\in\mathcal{I}_N}\langle\psi_I\rangle_G^2 \le \frac{1}{q}\sum_{\substack{S\subseteq[N]\\|S|=q-1}} 1 = \frac{1}{q}\binom{N}{q-1} \le \frac{N^{q-1}}{q!}.$$

From this and (14), we deduce that

$$\|\nabla W_N\|_2^2 \le \frac{\beta^2\sigma_N^2 N^{q-1}}{q!} = \frac{\beta^2}{q},$$

where the last equality follows from the definition of σ_N^2. This implies that $W_N(G)$ is $|\beta|/\sqrt{q}$-Lipschitz as a function of $G = (g_I)_{I\in\mathcal{I}_N}$. Hence, the Gaussian concentration inequality gives the desired result. □

Proof of Corollary **??**. The case $\beta = 0$ is trivial, so assume $\beta \neq 0$. Put

$$a := \frac{q}{2} - 1, \qquad W_N := \log Y_N(\beta).$$

It then holds that

$$\begin{aligned}\log Z_N - \mathbb{E}\log Z_N &= (\log Z_N - \log \mathbb{E}Z_N) + (\log \mathbb{E}Z_N - \mathbb{E}\log Z_N)\\ &= W_N - \mathbb{E}W_N\\ &= \big(W_N - (Y_N(\beta)-1)\big) + (Y_N(\beta)-1) - \mathbb{E}W_N,\end{aligned}$$

which yields

$$N^a(\log Z_N - \mathbb{E}\log Z_N) = N^a\big(W_N - (Y_N(\beta)-1)\big) + N^a(Y_N(\beta)-1) - N^a\mathbb{E}W_N.$$

By Theorem **??**, we have

$$N^a(Y_N(\beta)-1) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi. \tag{19}$$

Let $C_0 \in (0,\infty)$ be such that $|\log(1+x)-x| \le C_0x^2$ for all $x \in [-1/2, 1/2]$. Since $W_N = \log Y_N(\beta) = \log(1+(Y_N(\beta)-1))$, it follows that

$$\mathbb{E}\big[1\wedge(N^a|W_N-(Y_N(\beta)-1)|)\big] \le \mathbb{E}\big[1\wedge(C_0N^a|Y_N(\beta)-1|^2)\big] + \mathbb{P}(|Y_N(\beta)-1|>1/2).$$

By (19), $|Y_N(\beta)-1| \xrightarrow{\mathrm{p}} 0$ and $(N^a|Y_N(\beta)-1|)_{N\ge1}$ is tight. Thus, the right-hand side of the above inequality converges to 0 as $N\to\infty$. Hence, $N^a(W_N-(Y_N(\beta)-1)) \xrightarrow{\mathrm{p}} 0$. Therefore, it suffices to show that $\mathbb{E}W_N = o(N^{-a})$.

We first claim that

$$-\frac{\beta^2}{2q} \le \mathbb{E}W_N \le 0. \tag{20}$$

Indeed, the upper bound follows from Jensen's inequality, while the lower bound follows from (13) with $\lambda = 1$ and the identity $\mathbb{E}e^{W_N} = 1$. By Theorem **??**,

$$\delta_N^2 := \mathbb{E}(Y_N(\beta) - 1)^2 = O(N^{-(q-2)}). \tag{21}$$

Henceforth, we consider sufficiently large N so that $\delta_N \le e^{-1}$. Since $e^{W_N} - 1 = Y_N(\beta) - 1$, Markov's inequality implies that for any $t \ge 1$,

$$\mathbb{P}(W_N < -t) \le \frac{\delta_N^2}{(1 - e^{-t})^2} \le C\delta_N^2. \tag{22}$$

On the other hand, by $-\mathbb{E}W_N \le K_\beta := \beta^2/(2q)$ due to (20) and (12), for any $t > K_\beta$,

$$\mathbb{P}(W_N < -t) \le \exp\left(-\frac{q(t - K_\beta)^2}{2\beta^2}\right). \tag{23}$$

Set

$$T_N := \max\left\{1,\, K_\beta + \frac{2|\beta|}{\sqrt{q}}\sqrt{\log(1/\delta_N)}\right\}.$$

We represent

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] = \mathbb{P}(W_N < -1) + \int_1^\infty \mathbb{P}(W_N < -t)\,\mathrm{d}t.$$

Applying (22) to the first term and the integral over $[1, T_N]$ and applying (23) to the integral over (T_N, ∞), we obtain

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le CT_N\delta_N^2 + \int_{T_N}^\infty \exp\left(-\frac{q(t - K_\beta)^2}{2\beta^2}\right)\mathrm{d}t.$$

By the Gaussian tail bound $\int_a^\infty e^{-t^2}\mathrm{d}t \le Ce^{-a^2}$ for any $a \ge 0$, the second term is bounded by $C\delta_N^2$. Consequently,

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(1/\delta_N)}. \tag{24}$$

Now set $h(w) := e^w - 1 - w$. Note that

$$h(w) \le C(e^w - 1)^2 \quad (w \ge -1), \qquad h(w) \le |w| \quad (w < -1).$$

This, combined with $e^{W_N} - 1 = Y_N(\beta) - 1$ and (24), yields

$$\mathbb{E}h(W_N) \le C\mathbb{E}(Y_N(\beta) - 1)^2 + \mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(1/\delta_N)}.$$

Moreover, since $h(W_N) = Y_N(\beta) - 1 - W_N$ and $\mathbb{E}(Y_N(\beta) - 1) = 0$, we have $-\mathbb{E}W_N = \mathbb{E}h(W_N)$. Therefore, by (20), we deduce that

$$0 \le -\mathbb{E}W_N \le C\delta_N^2\sqrt{\log(1/\delta_N)}.$$

From this and (21), it follows that

$$\mathbb{E}W_N = O\Big(N^{2-q}\sqrt{\log N}\Big) = o(N^{-(q/2-1)}),$$

which completes the proof. □